

tions on that circuit are located on only one side of the substrate, except a very small crossover connection which uses the second side of the substrate.

Some situations where double-sided flex is used are:

1. When circuit density and layout cannot be routed on a single layer
2. Ground and power plane applications
3. For shielding applications
4. For dense surface-mount assembly

Salient features of the double-sided flex are:

1. Component assembly available on both sides
2. Two conductive layers
3. Fingers are an integral part of the conductor patterns

Multilayer flex circuits

Flex circuits having three or more layers of conductors are known as multilayer flex circuits. Commonly, the layers are interconnected by means of plated-through holes, though this is not a requirement by definition; it is possible to provide openings to access lower circuit level features.

The layers of a multilayer flex circuit may or may not be continuously laminated together throughout the construction, with the obvious exception of the areas occupied by plated-through holes. The practice of discontinuous lamination is common in cases where maximum flexibility is required. This is accomplished by leaving unbonded the areas where flexing or bending is to occur.

Multilayer flex circuits are used when circuit density and layout cannot be routed on a single layer, and for:

1. Ground and power plane applications
2. Shielding applications
3. Dense surface-mount assembly
4. Increased circuit density
5. EMI/RFI shielding

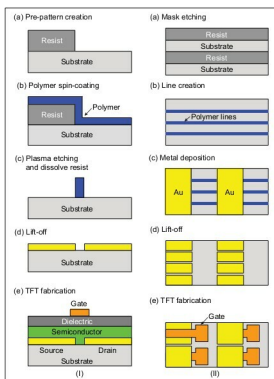


Fig. 8: Fabrication of short-channel transistors using spin-coating-induced edge effect (I, side view) and the specific process used for TFT fabrication (II, top view)

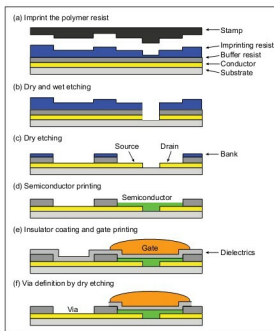


Fig. 9: TFT fabrication process by self-aligned, one-step, multilayered patterning

6. Controlled impedance with shielding

Multilayer flex circuits are used in high-density SMT electronic applications, and such sectors as automotive,

aerospace, medical, and test and measurement.

Rigid-flex circuits

Rigid-flex circuits are a hybrid construction with rigid and flexible substrates which are laminated together into a single structure. Rigid-flex circuits should not be confused with rigidised flex constructions that are simply flex circuits to which a stiffener is attached to support the weight of the electronic components locally. A rigidised or stiffened flex circuit can have one or more conductor layers. Thus, while the two terms may sound similar, they represent products that are quite different.

The layers of a rigid flex are also normally electrically interconnected by means of plated-through holes. Over the years, rigid-flex circuits have enjoyed tremendous popularity among military product designers. However, the technology has found increased use in commercial products. While often considered a specialty product for low-volume applications because of the challenges, rigid-flex boards are normally multilayer structures, but two-metal-layer constructions are also sometimes used.

Benefits of rigid-flex circuits include:

1. Use of third dimension creates an optimal solution for applications with extreme space limitations
2. Replace bulky wire harnesses with a compact, yet robust design
3. Flexible assemblies reduce connectors as well as labour, yield, transmission and reliability issues
4. These hybrid circuits can have up to sixteen layers

Polymer thick-film flex circuits

Polymer thick-film (PTF) flex circuits are true printed circuits wherein the conductors are printed onto a polymer base film. They are typically single-conductor-layer structures. However,